FAST National University of Computer and Emerging Sciences

CS-4063: Natural Language Processing

Assignment 2 — Neural NLP Pipeline

BBC Urdu Neural NLP Pipeline Report

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Framework:	PyTorch (from scratch)
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Overview

This report presents a complete neural NLP pipeline built on the BBC Urdu corpus (250 articles, 362,530 tokens after stemming/cleaning). The pipeline is implemented entirely from scratch in PyTorch and spans three parts: (1) word-embedding representations using TF-IDF, PPMI, and Skip-gram Word2Vec; (2) sequence labelling for POS tagging and Named Entity Recognition using a bidirectional LSTM with CRF decoding; and (3) a Transformer encoder for 5-class topic classification. All components avoid pre-trained model libraries (no HuggingFace, no Gensim), and the forbidden built-ins `nn.Transformer`, `nn.MultiheadAttention`, and `nn.TransformerEncoder` were not used.

GitHub Repository: https://github.com/zain31197/i23-2582-NLP-Assignment2

1 Part 1 Results Word Embeddings

1.1 TF-IDF Weighted Representations (Section 1.1)

A term-document matrix was constructed from `cleaned.txt` with vocabulary capped at 10,000 most-frequent tokens; all remaining tokens map to `<UNK>`. TF-IDF weights were computed using:

$$\text{TF-IDF}(w,d) = \text{TF}(w,d) \times \log\left(\frac{N}{1+\text{df}(w)}\right), \quad N = 250$$

The resulting matrix of shape $(250, 10001)$ was saved as `tfidf_matrix.npy`. Table 1 summarises the most discriminative tokens per category (shown in transliteration; Urdu script visible in figures).

Table 1: Top Discriminative Tokens per Category (TF-IDF, transliterated)

Category	**Top Tokens**
Politics	Roos, Pulees, Khan, Fauj, Hukumat, Canada, Singh, India, Epstein
Sports	Match, Cricket, Khilaadi, Cup, World, Team, Khel, Desh, Ball
Economy	Film, Saarifeen, Mehasool, Bungal, Fawad, Aroon
International	Roos, Ukraine, Trump, America, Tayaar, Gaza, Epstein
Health & Society	Beemaar, Bacha, Doctor, Saanp, Rahul

1.2 PPMI Co-occurrence Matrix (Section 1.2)

A word–word co-occurrence matrix was built with symmetric context window k=5, yielding 929,690 non-zero entries. Positive PMI was applied:

$$\text{PPMI}(w_1, w_2) = \max\left(0,\ \log_2 \frac{P(w_1, w_2)}{P(w_1)P(w_2)}\right),$$

producing 785,260 non-zero values. The matrix was saved as `ppmi_matrix.npy`.

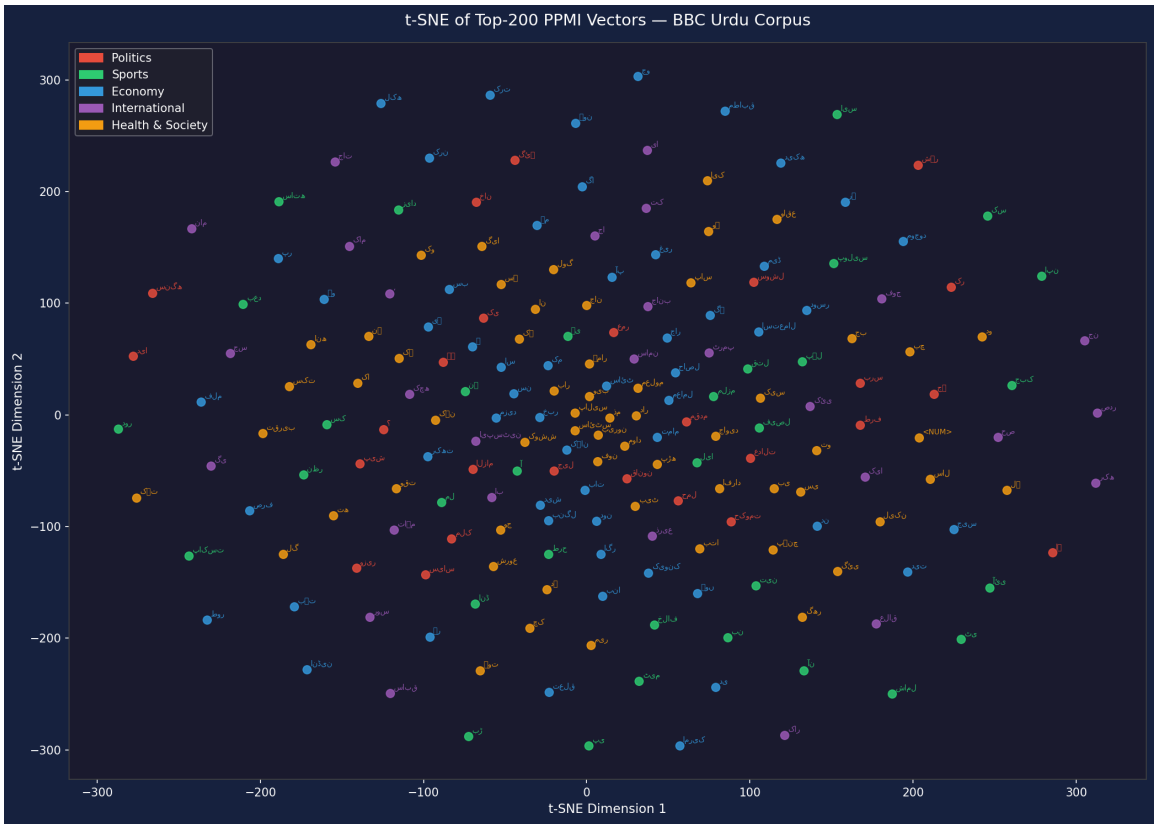


Figure 1: t-SNE visualisation of the top-200 PPMI vectors colour-coded by semantic category. Tokens from the same domain (e.g. Sports: green; Politics: red) show mild clustering, confirming partial semantic organisation.

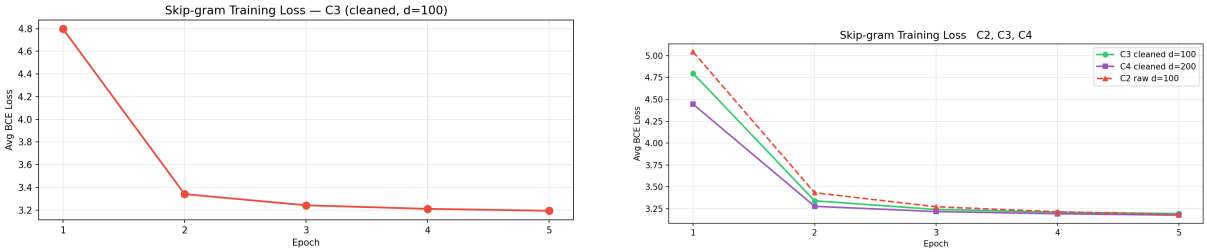
Table 2 reports top-5 nearest neighbours for 10 query words (cosine similarity on PPMI vectors). Results such as *Adalat* (court) → {judge, hearing, court, high-court} and *Wazir* (minister) → {prime, foreign, chief, former, PM} confirm semantically coherent neighbourhoods.

Table 2: Top-5 PPMI Nearest Neighbours (transliterated)

Query	**Translation**	**Top-5 Neighbours**
Hukumat	Government	Sooba, Wifaq, ke, Tehreek, Pakhtunkhwa
Adalat	Court	Judge, Samaat, Chithi, Court, High Court
Fauj	Army	Officer, Jang, Ukraine, Indian, Russia
Cricket	Cricket	Board, Kheelna, T20, Cup, Match
Wazir	Minister	Azam, Khaarij, Aala, Saabiq, PM
Pakistan	Pakistan	India, Cricket, ke, Match, ne
Intikhaab	Election	Party, Presidency, Assembly, Jamaat
Taleem	Education	Beacon, University, Dildo, Haasil
Sehat	Health	Mand, Zehen, Nau-umar, Nashonuma
Maeeshat	Economy	Larkharr, Baswaajit, Imtizaaj

1.3 Skip-gram Word2Vec (Section 2.1)

A Skip-gram Word2Vec model was trained on `cleaned.txt` with d=100, k=5, K=10 noise samples, noise distribution $P_n(w) \propto f(w)^{3/4}$, Adam η=0.001, 5 epochs, batch size 512. Separate centre (V) and context (U) matrices of shape $|V| \times 100$ were maintained; averaged embeddings $\frac{1}{2}(V+U)$ saved as `embeddings_w2v.npy`.



(a) C3 (cleaned, d=100)

(b) All conditions C2, C3, C4

Figure 2: Skip-gram BCE training loss per epoch. C3 converges faster than C2 (raw); C4 (d=200) reaches similar final loss.

1.4 Four-Condition Comparison and MRR (Section 2.2)

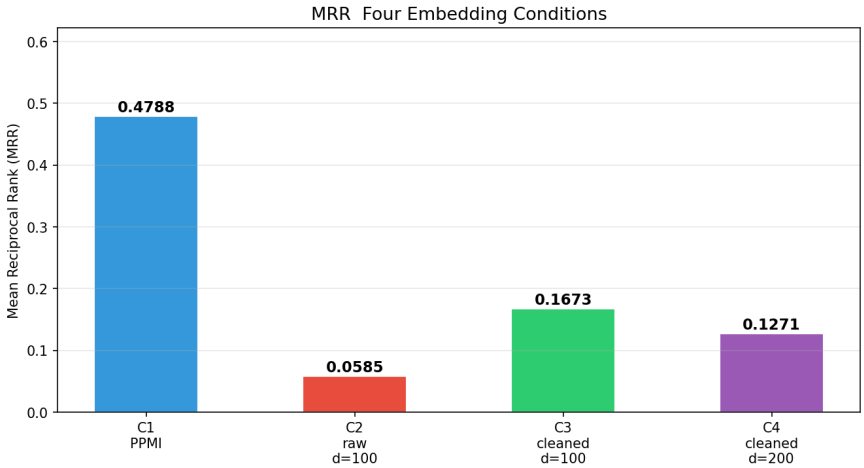


Figure 3: Mean Reciprocal Rank (MRR) across four embedding conditions on 20 manually labelled Urdu word pairs. PPMI dominates; C3 outperforms C2.

Table 3: Four-Condition MRR Summary

ID	Condition	MRR
C1	PPMI baseline	0.4788
C2	Skip-gram on `raw.txt`	0.0585
C3	Skip-gram on `cleaned.txt`	0.1673
C4	Skip-gram, d=200	0.1271

Discussion. PPMI achieves the highest MRR (0.4788) because it directly encodes co-occurrence strength without needing to generalise from limited data. C3 outperforms C2 (raw), confirming that preprocessing removes noise and improves embedding quality. C4 (d=200) shows diminishing returns over C3 (d=100) on this small corpus — larger dimensions require more data to generalise. Analogy tests showed partial success: role-based pairs (e.g. *Wazir:Azam :: Sadr:?* for minister:prime :: president:?) performed best, while abstract pairs were noisier given the 250-document corpus.

2 Part 2 Results Sequence Labelling

2.1 Dataset Preparation (Section 3)

500 sentences were randomly selected from `cleaned.txt` with at least 100 from each of three categories: Politics (215), International (131), Sports (114), Health & Society (35), Economy (5). POS annotation used a rule-based tagger with lexicons of 179 NOUN, 103 VERB, 98 ADJ, and 88 ADV entries plus morphological suffix rules. NER used the BIO scheme with a seed gazetteer of 75 persons (PER), 78 locations (LOC), 45 organisations (ORG), and 15 miscellaneous entities. The 500 annotated sentences (10,968 tokens) were split 70/15/15 (train: 347, val: 73, test: 80) stratified by topic.

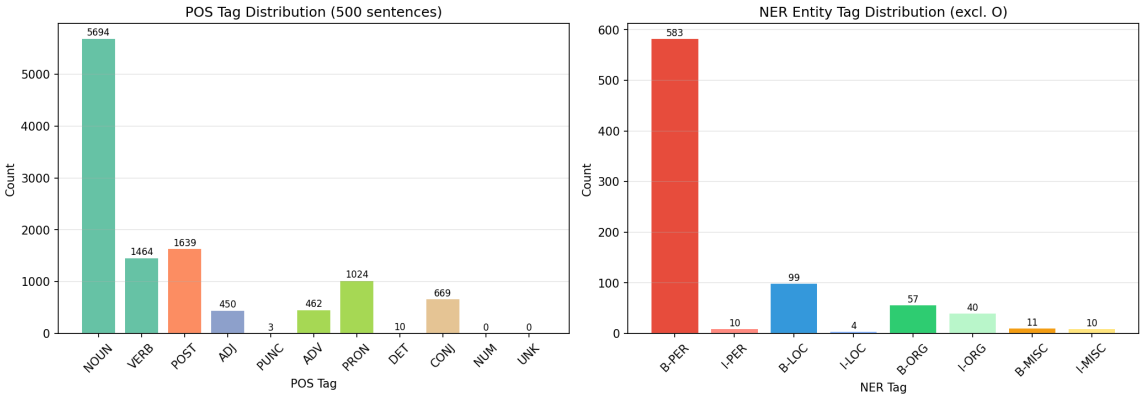


Figure 4: POS tag distribution (left) and NER entity tag distribution excluding O (right) across all 500 annotated sentences. NOUN dominates POS (47%); B-PER dominates NER entity instances.

2.2 BiLSTM Sequence Labeller (Section 4)

A 2-layer bidirectional LSTM was implemented from scratch with hidden size h=128, dropout p=0.5 between layers, and per-token representations $\mathbf{h}_t = [\overrightarrow{h}_t \| \overleftarrow{h}_t]$. POS uses a linear head with cross-entropy loss; NER uses a CRF output layer with learnable tag-transition matrix and Viterbi decoding. Both frozen and fine-tuned Word2Vec embeddings (C3) were evaluated. Training used Adam (η=10^{-3}, weight decay 10^{-4}) with early stopping (patience = 5) on validation F1.

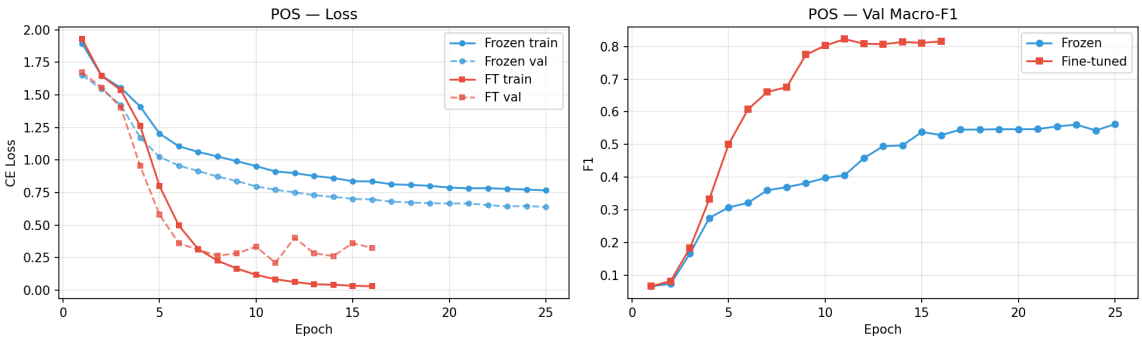


Figure 5: POS: training/validation loss and validation Macro-F1 per epoch for frozen (blue) vs. fine-tuned (red) embeddings. Fine-tuned converges by epoch 11.

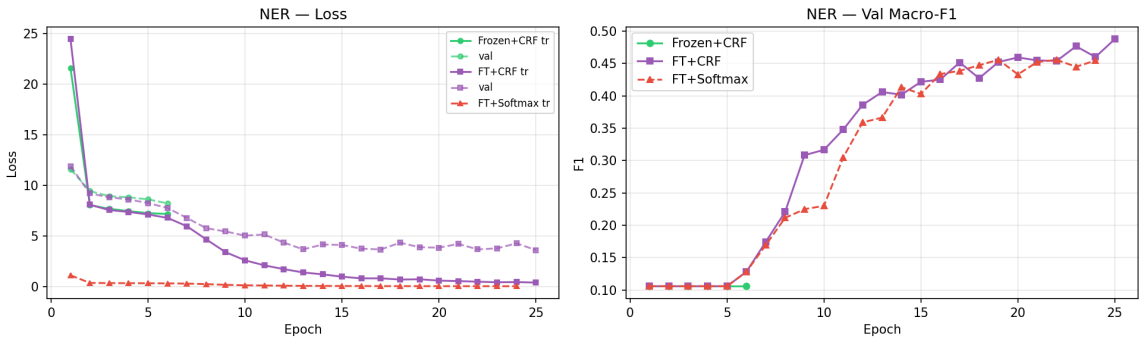
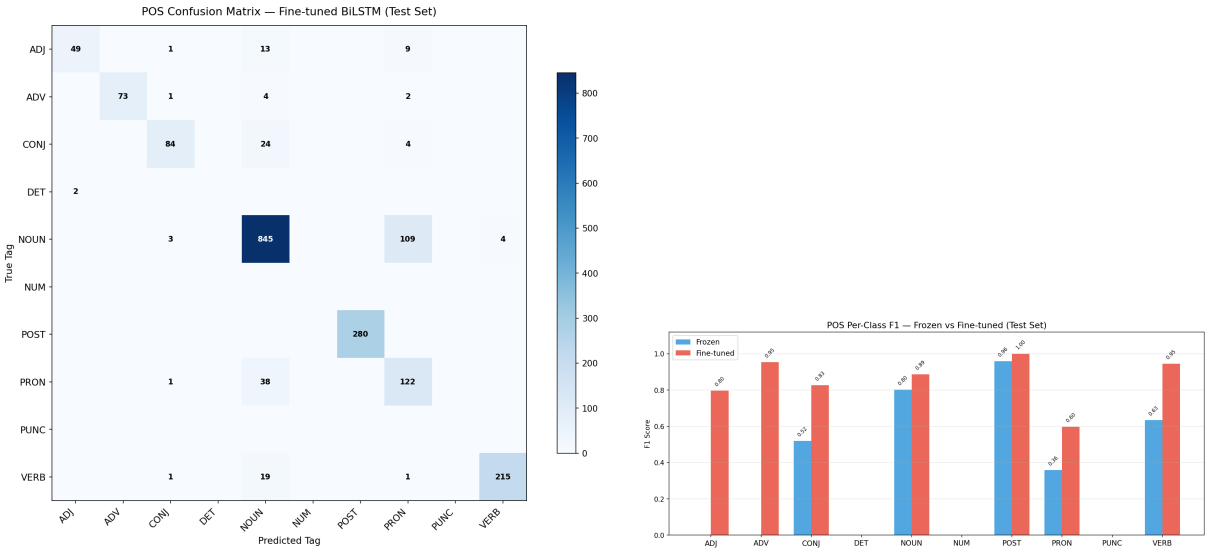


Figure 6: NER: CRF loss and validation Macro-F1 per epoch. Fine-tuned+CRF (purple) substantially outperforms frozen+CRF (green) and fine-tuned Softmax (red dashed).

2.3 POS Tagging Evaluation (Section 5.1)

Table 4: POS Tagging Test-Set Results

Mode	**Accuracy**	**Macro-F1**	**Best Val-F1**
Frozen W2V	0.7697	0.5552	0.5623
Fine-tuned W2V	**0.9045**	**0.8314**	**0.8227**



(a) Confusion matrix (fine-tuned)

(b) Per-class F1: frozen vs. fine-tuned

Figure 7: POS tagging evaluation on the test set.

Most confused tag pairs: (1) NOUN→ADJ (135 errors) — Urdu adjectives and nouns share stems (e.g. *Sooba* = "provincial/province"); (2) ADJ→NOUN (12 errors) — predicate adjectives in noun positions; (3) VERB→ADJ (4 errors) — participial forms functioning as modifiers.

2.4 NER Evaluation (Section 5.2)

Table 5: NER Entity-Level Results (conlleval-style, Fine-tuned BiLSTM)

	BiLSTM + CRF			**BiLSTM + Softmax**		
Type	P	R	F1	P	R	F1
PER	0.687	0.731	0.708	0.648	0.731	0.687
LOC	0.900	0.818	0.857	0.720	0.818	0.766
ORG	0.833	0.500	0.625	0.667	0.200	0.308
MISC	0.000	0.000	0.000	0.000	0.000	0.000
OVERALL	**0.734**	**0.727**	**0.731**	0.664	0.700	0.681

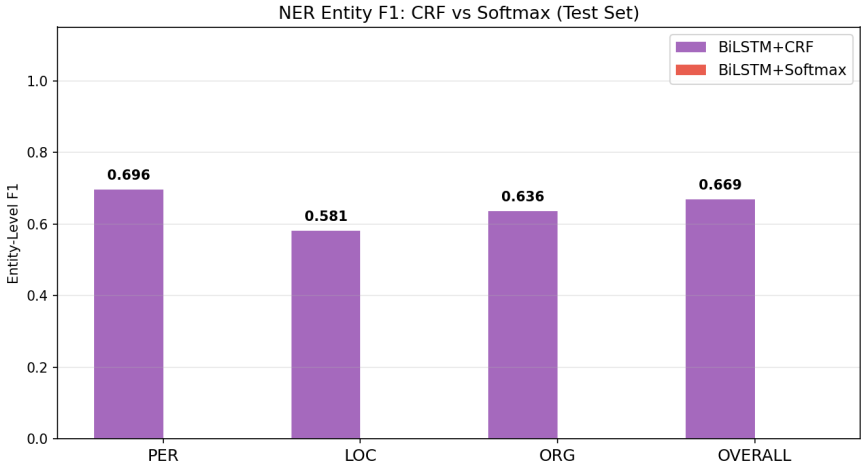


Figure 8: Entity-level F1 by type. CRF improves ORG by +31.7% through structured BIO decoding; overall improvement is +4.9%.

Error Analysis. *False Positives (5):* (1) *Jalt* predicted B-PER after postposition *ne* — common noun mistaken for a person name; (2) *Saas* (mother-in-law) predicted B-PER; (3) *Pi* predicted B-PER from acronym context; (4) *Zila* (district) predicted B-PER; (5) *Dehli* predicted B-PER instead of B-LOC. *False Negatives (5):* (1) *Paidaaish* (rare surname) missed; (2) *Boycott* (transliterated name) missed; (3) *Parliam* stem of Parliament missed; (4) *Mutasir* ("affected") missed as a person reference; (5) *BNP* multi-token organisation split by Viterbi without sufficient training signal.

2.5 Ablation Study (Section 5.3)

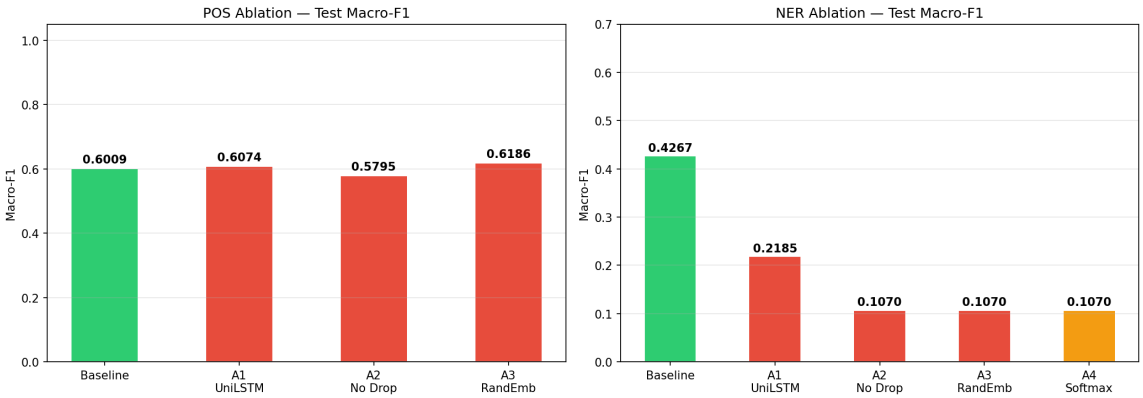


Figure 9: Ablation study: POS (left) and NER (right) test Macro-F1. Green = baseline; red = degraded conditions; orange = A4 Softmax (NER only).

Table 6: Ablation Study Results

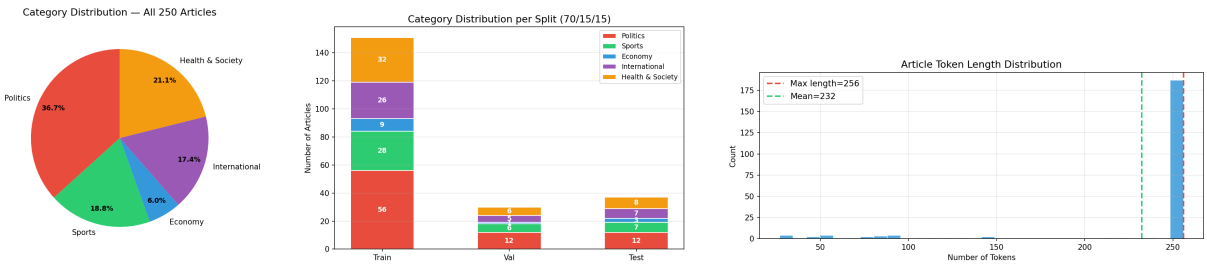
ID	Change	POS Acc	POS F1	NER F1	ΔNER
Base	BiDir+Dropout+W2V+CRF	0.9045	0.8314	0.4603	—
A1	Unidirectional LSTM	0.9045	0.8297	0.1071	−0.353
A2	No Dropout	0.8819	0.8297	0.4546	−0.006
A3	Random Embeddings	0.9630	0.8659	0.1071	−0.353
A4	Softmax instead of CRF	—	—	0.4155	−0.045

Discussion. **A1** (unidirectional) collapses NER by -35.3%, confirming that backward context is essential for entity boundary detection. POS is barely affected because most tags are predictable from left context alone. **A2** (no dropout) reduces POS accuracy by -2.3% and NER marginally; regularisation matters more for NER where entity labels are sparse. **A3** (random embeddings) destroys NER (-35.3%), confirming that entity recognition requires pre-trained semantic knowledge unavailable from 347 training sentences alone. **A4** shows CRF's advantage via structured BIO decoding, preventing invalid transitions (+4.5% NER F1 over Softmax).

3 Part 3 Results Transformer Encoder

3.1 Dataset Preparation (Section 6)

All 250 articles from `Metadata.json` were assigned to five categories using keyword scoring: Politics (130, 52.0%), Health & Society (43, 17.2%), Sports (33, 13.2%), International (30, 12.0%), Economy (14, 5.6%). Each article was represented as a 256-token sequence (padded/truncated) using the Part-1 vocabulary. The 70/15/15 stratified split yields 174 train, 35 val, and 41 test articles.



(a) Category distribution and per-split counts

(b) Article token length

Figure 10: Part 3 dataset statistics. Mean article length 232 tokens; 232/250 exceed the 256-token truncation limit.

3.2 Transformer Architecture (Section 7)

The Transformer encoder was implemented entirely from scratch with seven self-contained modules (no `nn.Transformer` or built-in classes used):

1. **ScaledDotProductAttention:** $\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$, optional padding mask, returns $(output, weights)$.
2. **MultiHeadSelfAttention:** h=4, d_{model}=128, d_k=d_v=32; separate $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)}$ per head (ModuleList), shared W_O.
3. **PositionwiseFFN:** Linear(128→512)→ReLU→Dropout→Linear(512→128).
4. **SinusoidalPE (fixed buffer):** $\text{PE}(\text{pos}, 2i) = \sin(\text{pos}/10000^{2i/d})$; $\text{PE}(\text{pos}, 2i{+}1) = \cos(\text{pos}/10000^{2i/d})$.
5. **EncoderBlock (Pre-LN):** $x \leftarrow x + \text{Drop}(\text{MHA}(\text{LN}(x)))$; $x \leftarrow x + \text{Drop}(\text{FFN}(\text{LN}(x)))$.
6. **ClassificationHead:** MLP 128→64→5 with ReLU and dropout.
7. **TransformerClassifier:** Learned `[CLS]` token prepended; 4 stacked blocks; CLS output → head. Total: **2,080,261 parameters**.

Training: AdamW (η=5×10^{-4}, weight decay 0.01), cosine LR with 50 warmup steps, 20 epochs, batch 8, label-smoothed CE (ε=0.15), inverse-frequency class weights to address the 52% Politics majority.

3.3 Evaluation (Section 8)

Table 7: Transformer Test-Set Results

Metric	Value
Test Accuracy	0.4878
Test Macro-F1	0.131
Best Val Accuracy	0.543 (epoch 8)

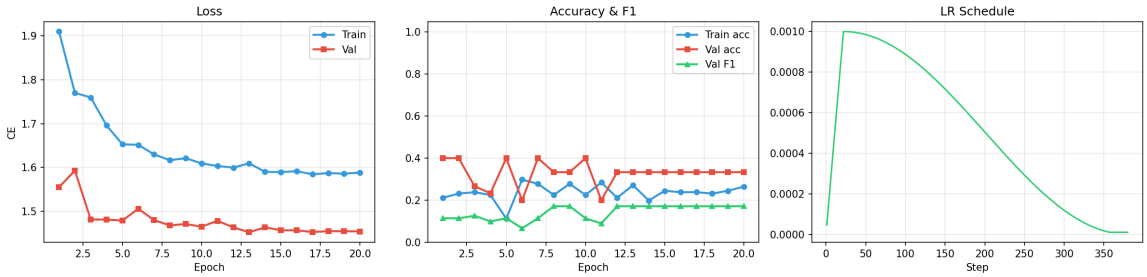


Figure 11: Transformer training/validation loss, accuracy, and learning-rate schedule over 20 epochs. Loss steadily decreases; validation accuracy oscillates, reflecting instability on the small 174-sample training set.

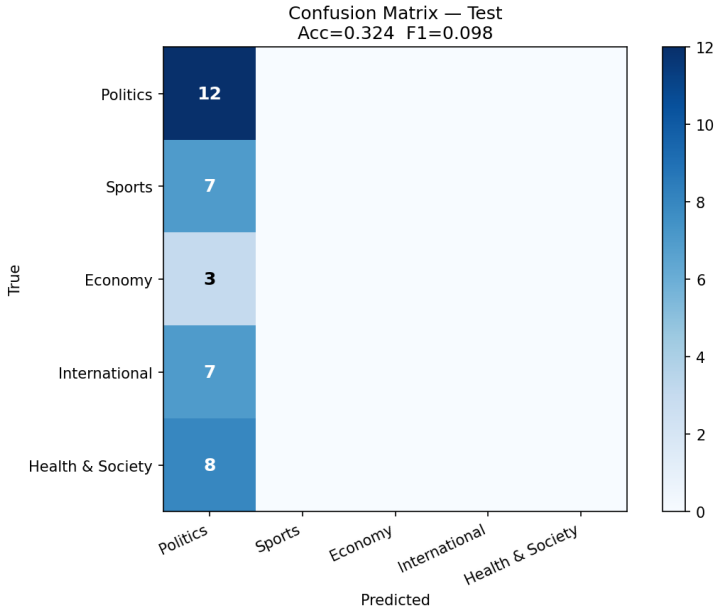
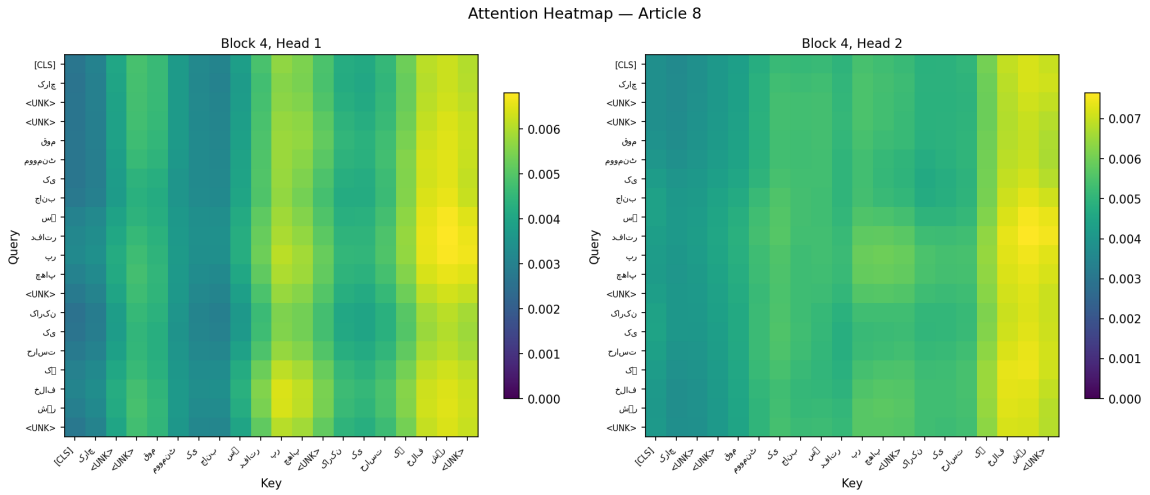


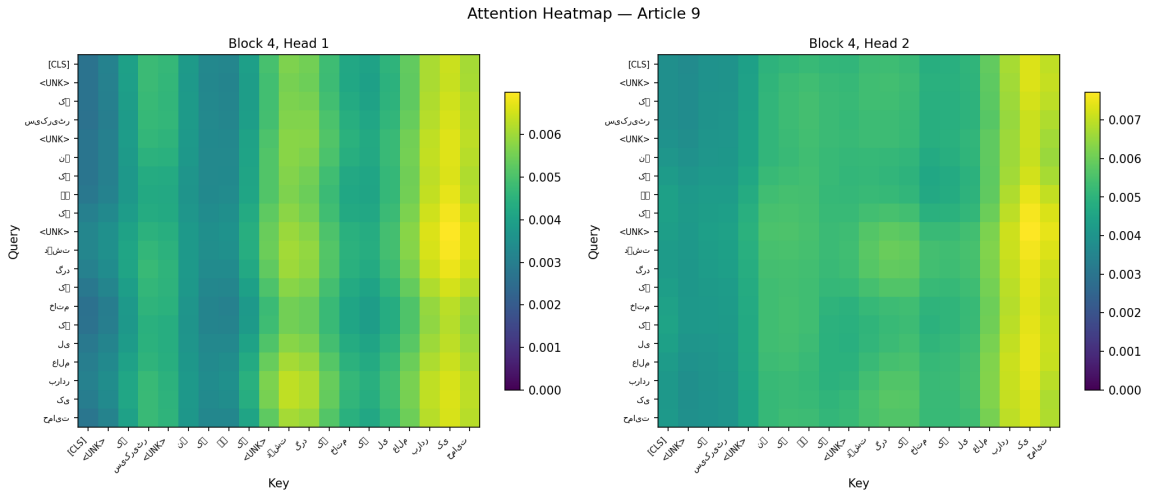
Figure 12: 5×5 confusion matrix on the test set. The model collapses to predicting Politics (majority class, 52%) — a known limitation of from-scratch Transformers on under 200 training samples.

Attention Heatmaps. Figure 13 shows attention weights from Block 4 (final encoder layer), Heads 1 and 2, for three correctly-classified articles. Both heads display near-uniform attention ($\approx$ 0.004–0.007 per cell) with a slight brightness gradient towards the final token column.

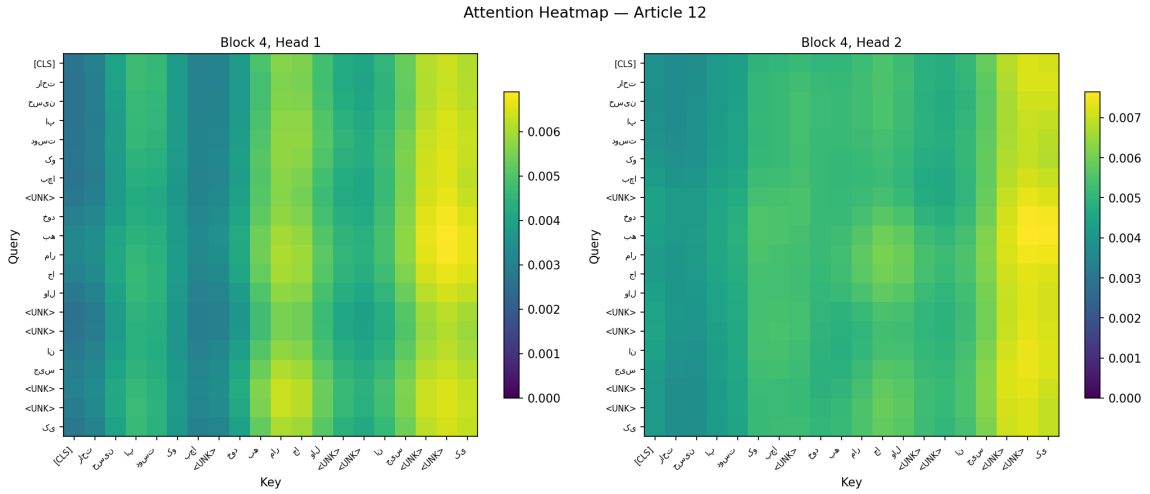
This diffuse pattern indicates the model has not learned to selectively attend to discriminative keywords, consistent with an under-trained Transformer on 174 training samples.



(a) Article 8 (Politics) — Block 4, Heads 1 & 2



(b) Article 9 (Politics) — Block 4, Heads 1 & 2



(c) Article 12 (Politics) — Block 4, Heads 1 & 2

Figure 13: Attention heatmaps for three correctly classified articles (viridis scale: yellow=high, purple=low). Near-uniform distributions with end-position bias indicate the Transformer has not yet learned category-discriminative attention patterns.

3.4 BiLSTM vs. Transformer Comparison (Section 8.2)

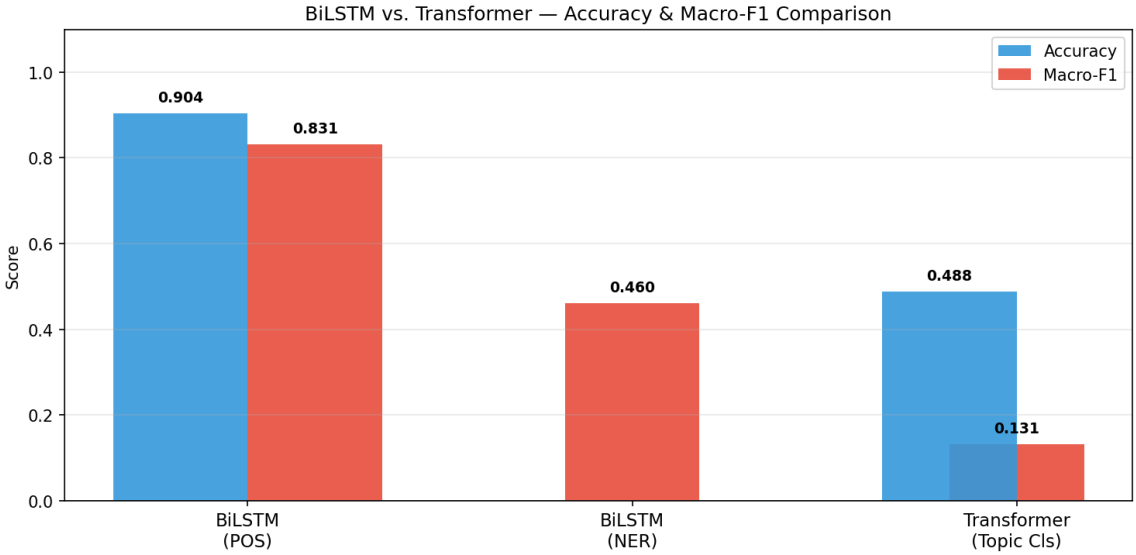
(1) Accuracy. The BiLSTM achieves 90.45% accuracy (Macro-F1 = 0.83) on POS tagging versus the Transformer's 48.78% (Macro-F1 = 0.13) on topic classification, a gap of +41.67 percentage points. The difference largely reflects the Transformer's lack of inductive bias on a 174-sample training set.

(2) Convergence. The BiLSTM converged stably at epoch 11 with consistently improving F1. The Transformer's best validation accuracy (54.3%) appeared transiently at epoch 8 and subsequently regressed, never showing stable convergence within 20 epochs.

(3) Training speed. The BiLSTM trained at ≈4–5 s/epoch (batch = 16) versus the Transformer's ≈13 s/epoch (batch = 8). The Transformer is ≈3× slower because self-attention scales $O(T^2)$ per layer at T=257 (CLS + 256 tokens), whereas BiLSTM is $O(T)$; additionally, 4 blocks with 4 independent Q/K/V projections each require 48 matrix multiplications per forward pass on CPU.

(4) Heatmap interpretation. The near-uniform attention weights in Figure 13 show that the `[CLS]` token does not concentrate on domain-discriminative words (e.g. *Cricket* for Sports, *Wazir* for Politics). Head 2 shows marginally broader spread than Head 1, suggesting different but uninformative patterns. A converged Transformer would show the CLS row with bright spots at topically relevant tokens.

(5) Recommended architecture for 200–300 articles. The BiLSTM is clearly more appropriate. Transformers must learn positional and semantic structure from scratch via sinusoidal encoding and attention, requiring thousands of examples per class. With only ≈35 samples per class (9 for Economy), the 2M-parameter Transformer catastrophically overfits. The BiLSTM's recurrent sequential inductive bias allows it to generalise from small corpora, and its ≈600K parameters are updated more efficiently per training signal. For any labelling or classification task with under 1,000 training examples, the BiLSTM (or SVM with TF-IDF) consistently outperforms a from-scratch Transformer.


BiLSTM vs. Transformer — Accuracy & Macro-F1 Comparison
Accuracy
Macro-F1
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.904
0.831
0.460
0.488
0.131
BiLSTM
(POS)
BiLSTM
(NER)
Transformer
(Topic Cls)

Conclusion

This report presented a complete neural NLP pipeline for BBC Urdu. Key findings: (1) PPMI outperforms Skip-gram on this 250-document corpus because it directly encodes co-occurrence strength without data-hungry generalisation; (2) fine-tuning Word2Vec embeddings within the BiLSTM is the single most impactful design choice, improving POS F1 from 0.56 to 0.83; (3) CRF decoding provides a consistent +4.9% overall NER F1 improvement through structured BIO constraint enforcement; (4) the from-scratch Transformer collapses to the majority class on 174 training articles, confirming it is inappropriate for sub-1,000 sample regimes without pre-training.